

New Collaborative Approach to Scientific Data Management with NOVA

**W. Mexner, E. Bründermann, M. Caselle, S. Funkner,
A. Kopmann, G. Niehues, N. Tan Jerome, M. Vogelgesang**

www.ibpt.kit.edu/wm

Institute for Beam Physics and Technology (IBPT)

49° 1' N - 8° 24' O



Karlsruhe Research Accelerator (KARA)

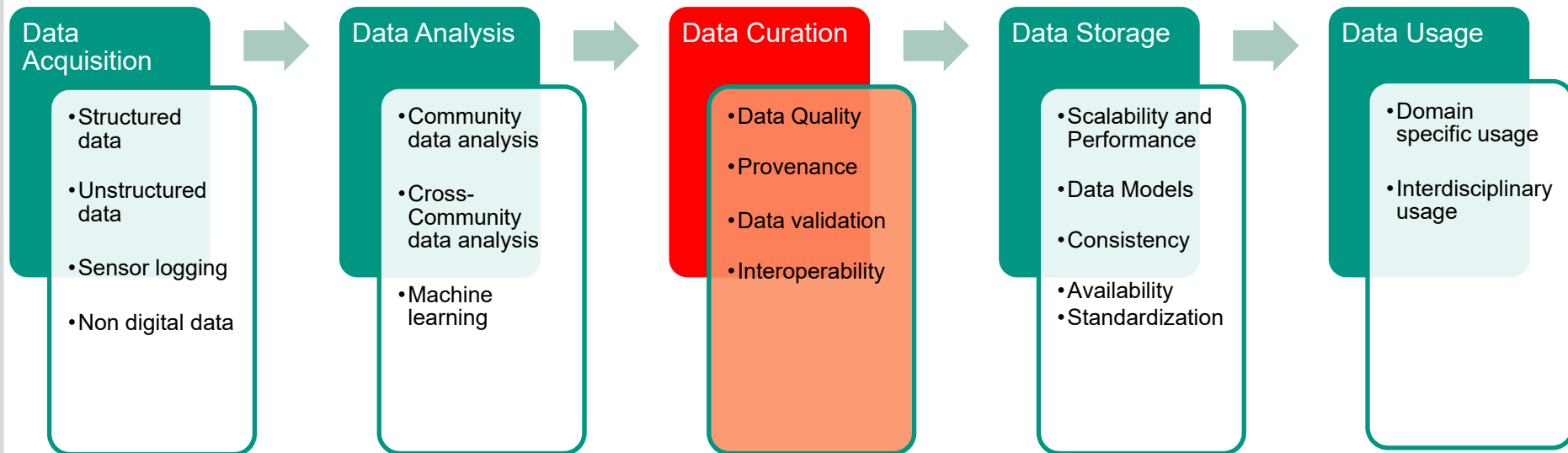
2.5 GeV electron storage ring, 110.4 m circumference

- Accelerator technology test facility
- Bunch length down to few ps
- Ultra precise beam diagnostic
- Synchrotron radiation source
- Ultra high speed tomographic station

How to manage long term TBytes of scientific data?
What's about open data?



Is your data useful for somebody in 10 years?



Adapted from Freitas et al.
New Horizons for a Data-Driven Economy
https://doi.org/10.1007/978-3-319-21569-3_6

Scientific data management: FAIR data principle

F indable

- Persistent ID
- Metadata online

A ccessible

- Data online
- Restrictions where needed

I nteroperable

- Use standards, controlled vocabs
- Common (open) formats

R eusable

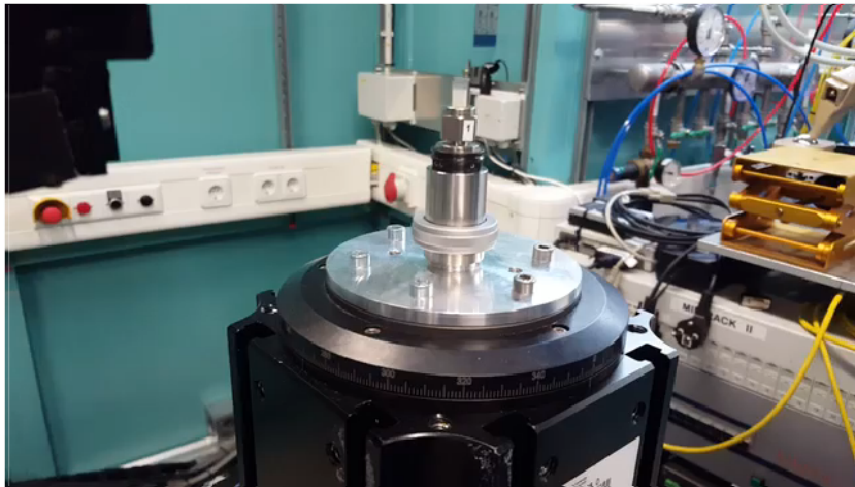
- Rich documentation
- Clear usage license

Wilkinson et al, Nature,
Scientific Data
doi:10.1038/sdata.2016.18

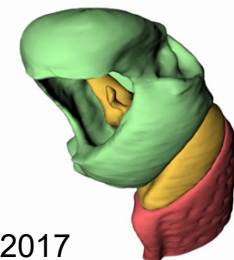
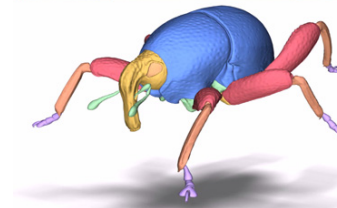
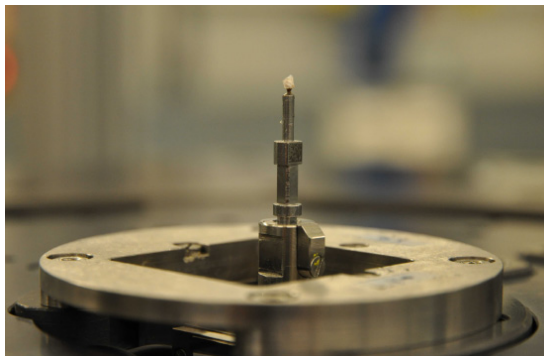
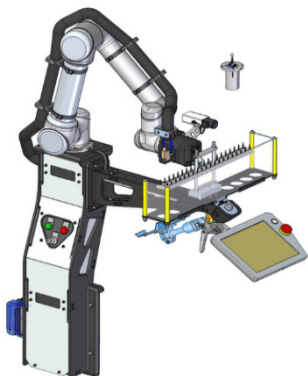
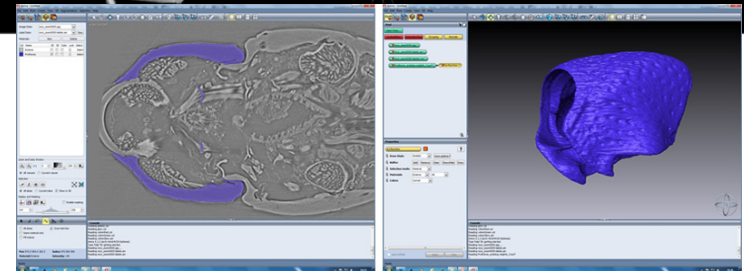
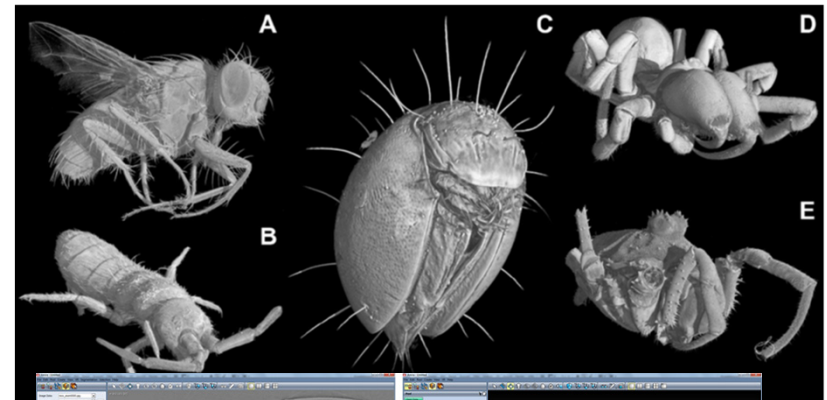
Ultra-fast high throughput X-ray imaging

90 seconds/tomogram
22 Tbyte/8h

Beamline automation



Segmentation for biologic analysis



Schmelzle et al., Proc.SPIE 2017

Challenges for high-throughput experiments

- **Huge amount of data (up to 22 TB/8h)**
 - Collaborators often do not have the necessary resources (computers, memory, software) available in order to process the tomograms at their home institutions

- **Data analysis is still very time-consuming (several weeks/tomogram)**
 - Segmentation of tomographic datasets is usually done manually

Bottleneck for processing of large collections

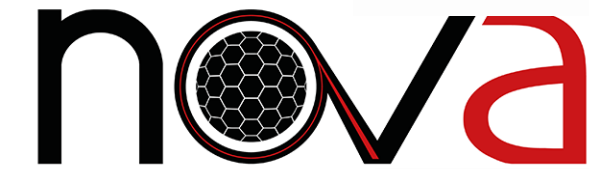
- **Results should be shared and open data**

Scientific data management for tomography

- Virtual analysis infrastructure for tomographic data @ KIT & DESY
- Remote access for external users
- Speedup of segmentation by an order of magnitude from weeks to hours (Biomedisa Algorithm)
- Joint analysis of selected datasets by specialized research groups



SPONSORED BY THE



SPONSORED BY THE



Friedrich-Schiller-Universität Jena

ERNST MORITZ ARNDT
UNIVERSITÄT GREIFSWALD

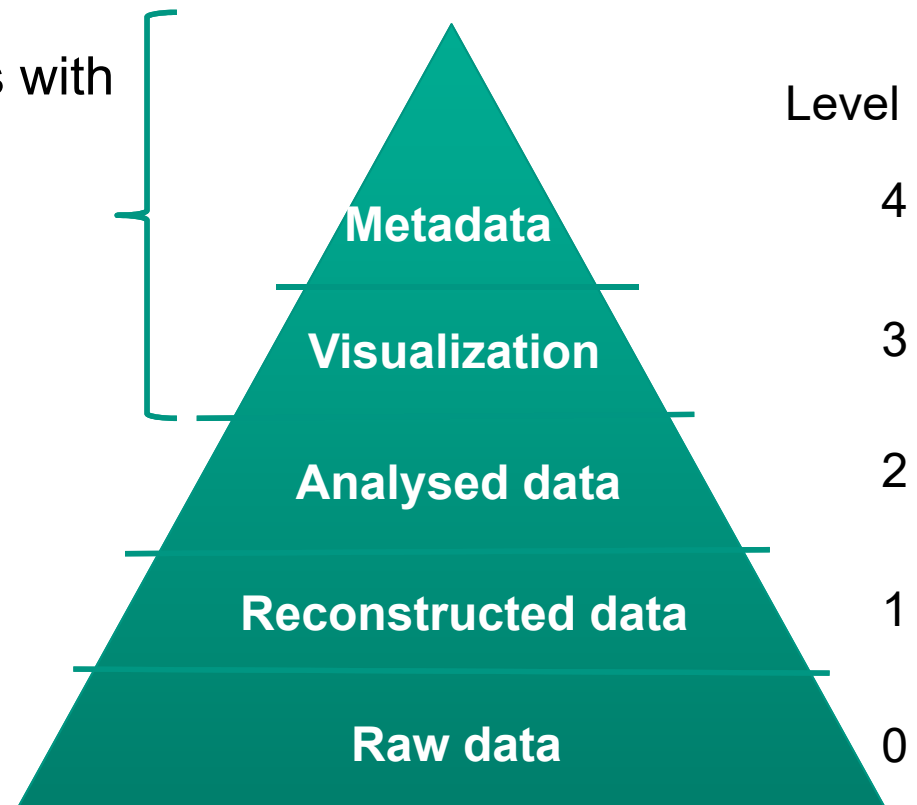


NOVA: Hierarchical organization of datasets

- **Long-term archive have long access times**
- Organize large dataset in levels with decreasing size is needed
- Standard: Only metadata is available in data catalogs

Solution:

- **Extend classical metadata by visualization data**
- **Developed 3D-visualization for data catalogs**





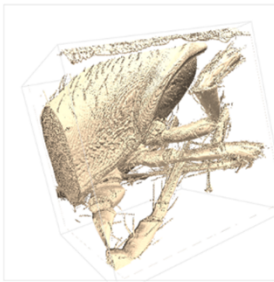
Datasets for **astor-2014**

- 
[matthias / Archie_03_local](#)
 Description of this dataset.
- 
[matthias / Archie_01](#)
- 
[matthias / Archie_02](#)
- 
[matthias / Archie_03](#)
- 
[matthias / Archie_04](#)
- 
[matthias / Archie_Nymph](#)
- 
[matthias / Belbe_01](#)
- 
[matthias / Belbe_02](#)
- 
[matthias / Eucrib_01](#)
- 
[matthias / Eucrib_02](#)
- 
[matthias / Eucrib_03](#)
- 
[matthias / Eucrib_04](#)
- 
[matthias / Eucrib_05](#)

NOVA - Network for Online Visualization and Synergistic Analysis of Tomographic Data
 A collaboration between Technical University of Darmstadt, University of Heidelberg, Karlsruhe Institute of Technology, Helmholtz Center Geesthacht, University of Tübingen, University of Jena and University of Greifswald.



Information - **astor-2014** / Gama_01



Gray value threshold
 94 255

Slicing
 X axis Y axis Z axis

Color mapping
 Gray Voids Jet

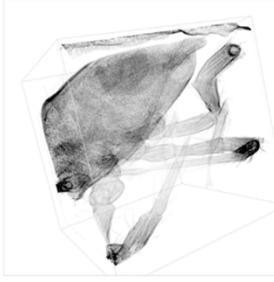
Render mode
 Voxel Surface

Zoom control
 0,0 1,0 0,0 1,0
 0,1 0,1
 x-y plane y-z plane

NOVA - Network for Online Visualization and Synergistic Analysis of Tomographic Data
 A collaboration between Technical University of Darmstadt, University of Heidelberg, Karlsruhe Institute of Technology, Helmholtz Center Geesthacht, University of Tübingen, University of Jena and University of Greifswald.



Information - **astor-2014** / Gama_01



Gray value threshold
 94 255

Slicing
 X axis Y axis Z axis

Color mapping
 Gray Voids Jet

Render mode
 Voxel Surface

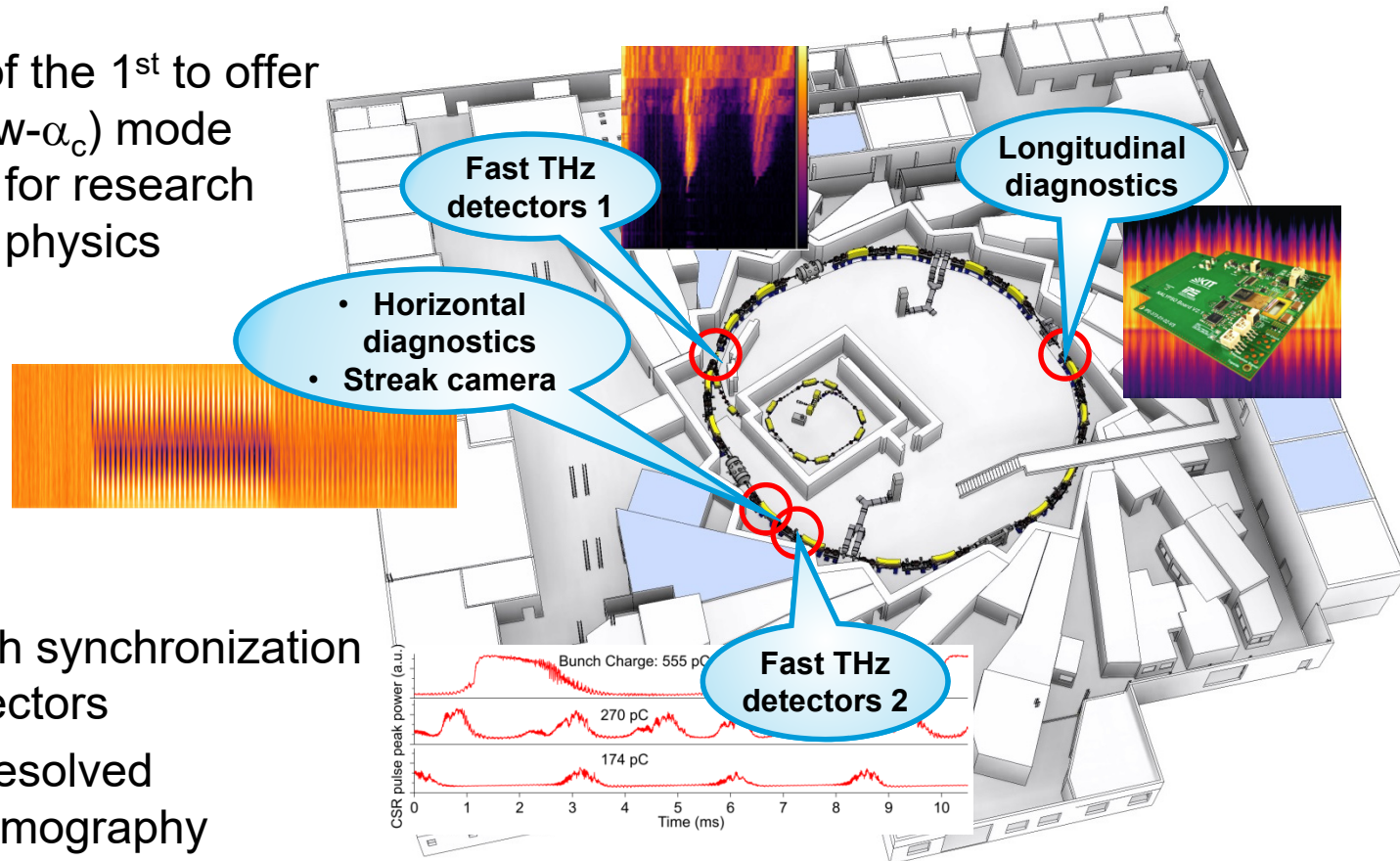
Zoom control
 0,0 1,0 0,0 1,0
 0,1 0,1
 x-y plane y-z plane

NOVA - Network for Online Visualization and Synergistic Analysis of Tomographic Data
 A collaboration between Technical University of Darmstadt, University of Heidelberg, Karlsruhe Institute of Technology, Helmholtz Center Geesthacht, University of Tübingen, University of Jena and University of Greifswald.

<https://ufo.kit.edu/>

Data management for accelerator diagnostics

- KIT: from one of the 1st to offer short bunch (low- α_c) mode to a test facility for research on short bunch physics



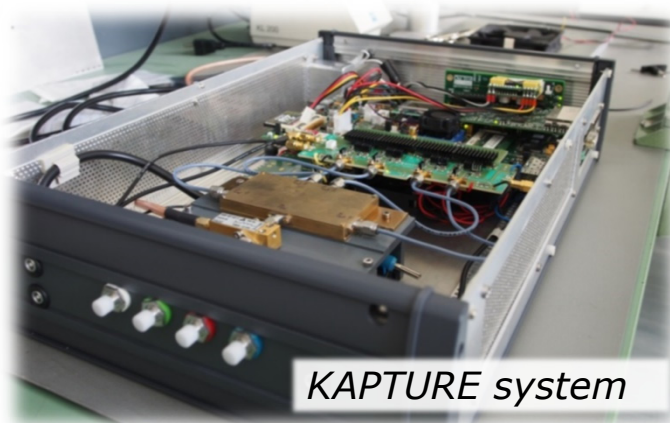
- Bunch-by-bunch synchronization of different detectors
- Towards time resolved phase space tomography

B. Kehrer et al. Simultaneous Detection of Longitudinal and Transverse Bunch Signals at a Storage Ring
arXiv:1709.08973v3 [physics.acc-ph]

New detectors for electron beam diagnostics

More realistic situation for data management:
Different data sources with different data structures

- ➔ **KAPTURE** (~ 4 GByte/s)
Continuous bunch by bunch diagnostics with pico second resolution by non-aquidistant programmable sampling.
- ➔ **KALYPSO** (~ 5 GByte/s)
Ultrafast line camera for femtosecond resolution through EO-modulation



KAPTURE system



KALYPSO system

M. Caselle et al., JINST 12 C03015 (2017)
L. Rota, M. Caselle et al., IBIC WEPG46 (2016)

Ongoing: Import data to NOVA

Publishing „FAIR“ machine diagnostics data:

Analysis of the actual situation:

- Data stored as HDF5 container in structured file system
- Machine parameters stored in NoSQL cassandra database
- Detector setup stored in elog-Book
- Important information also in paper-Logbooks

⇒ Introduction of data management rules required

⇒ Are these rules accepted?

How to motivate users for data management?

Wildcard based data set(s) selector

Static Data to be imported

Call specific metadata import tool
+List of items to be processed

Call GUI input tool
+List of items to be processed

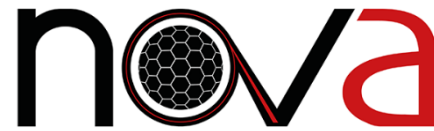
Generate metadata file for dataset(s)
Inject data to metadata catalogue
Move data safe to next storage tier

Offer automatic tools for data collection

Idea:

Adding metadata to data is more comfortable than manual data handling

Summary



BMBF grant 05K16VBK

- NOVA Portal introduced **scientific data management** to **tomographic data** @ KIT & DESY
- We are introducing **scientific data management** to **accelerator beam diagnostics** @ KIT with NOVA
- We are in close collaboration with accelerator scientists to establish a **new workflow**
- Long term **scientific data management** requires **data curation**